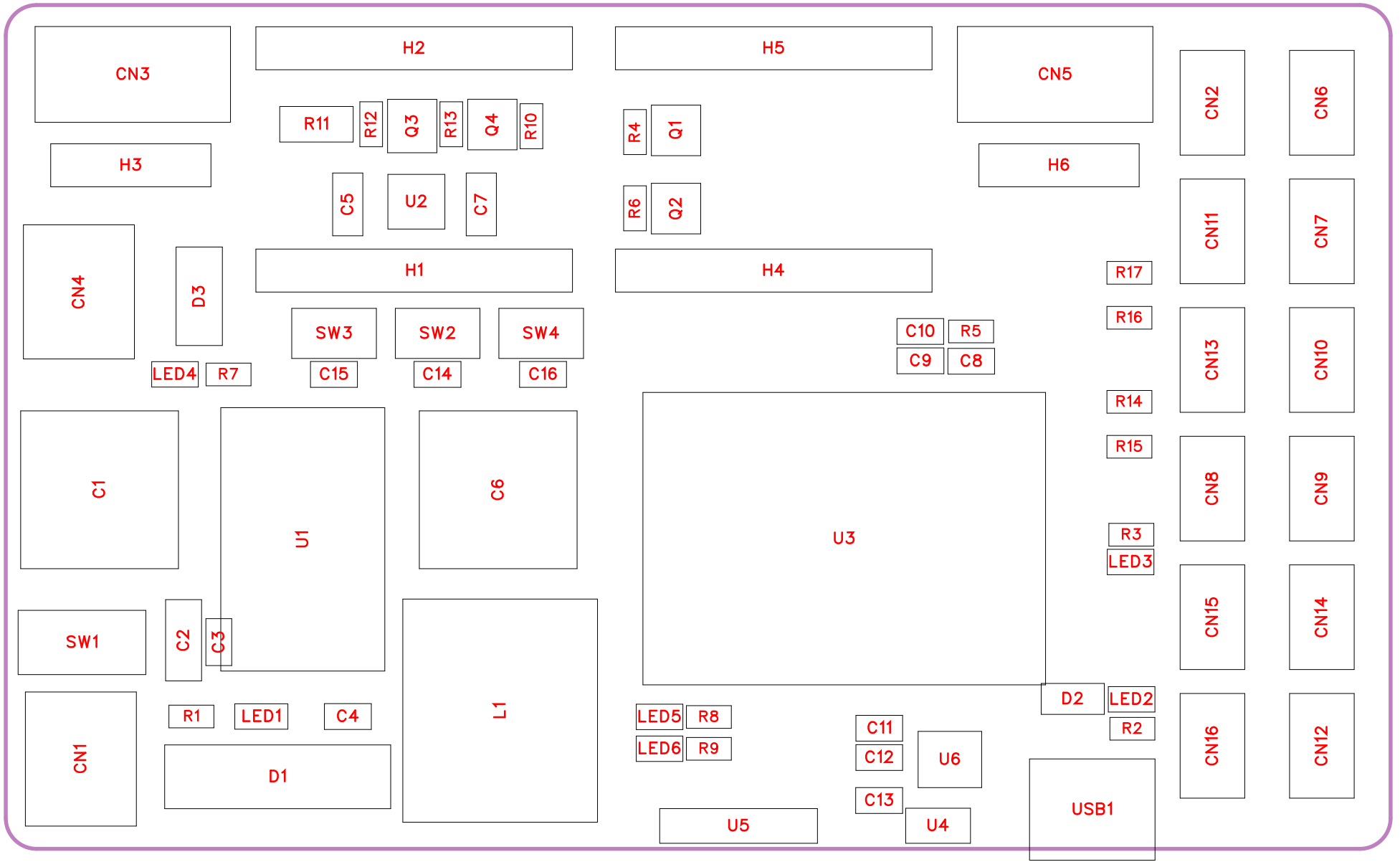
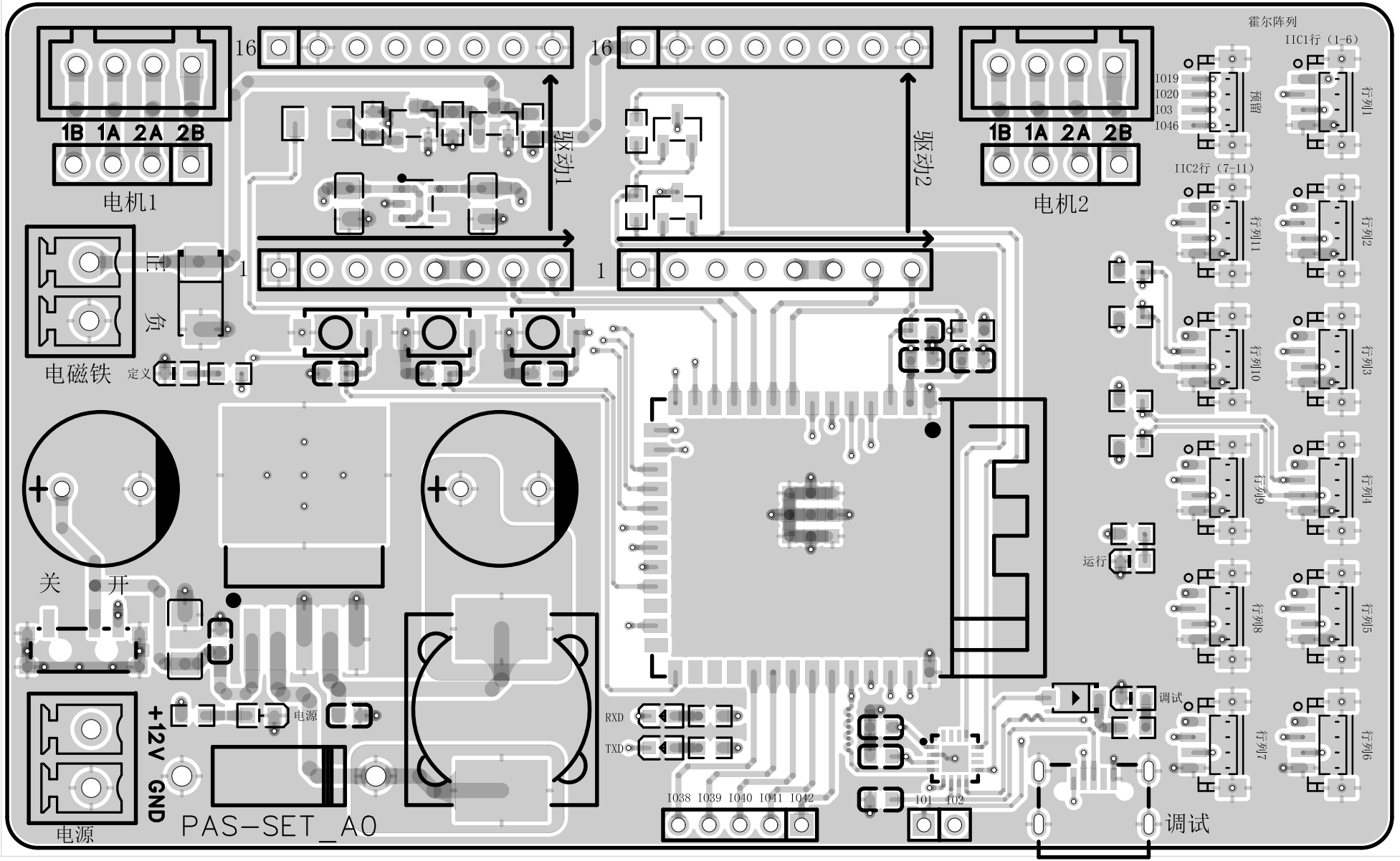
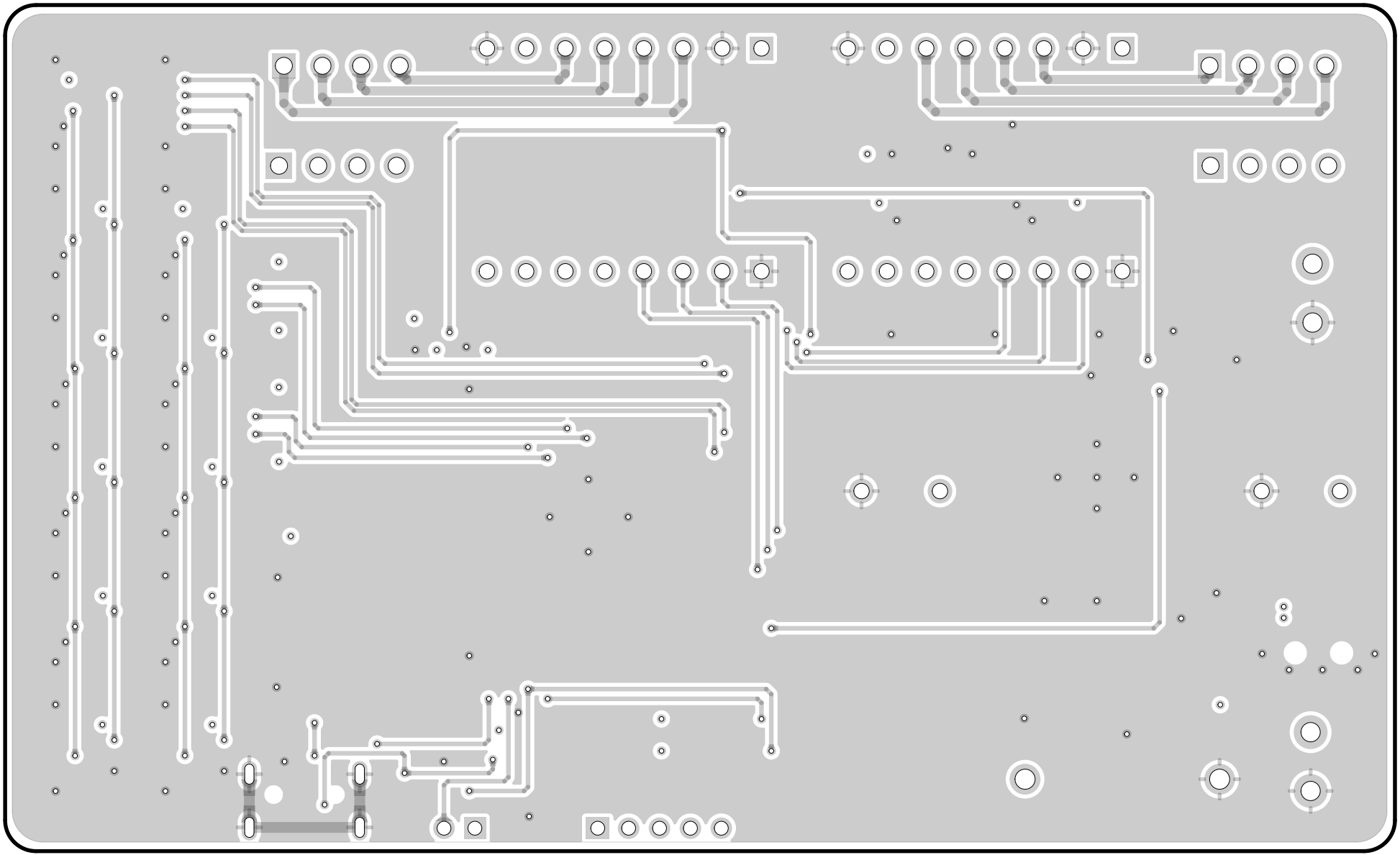
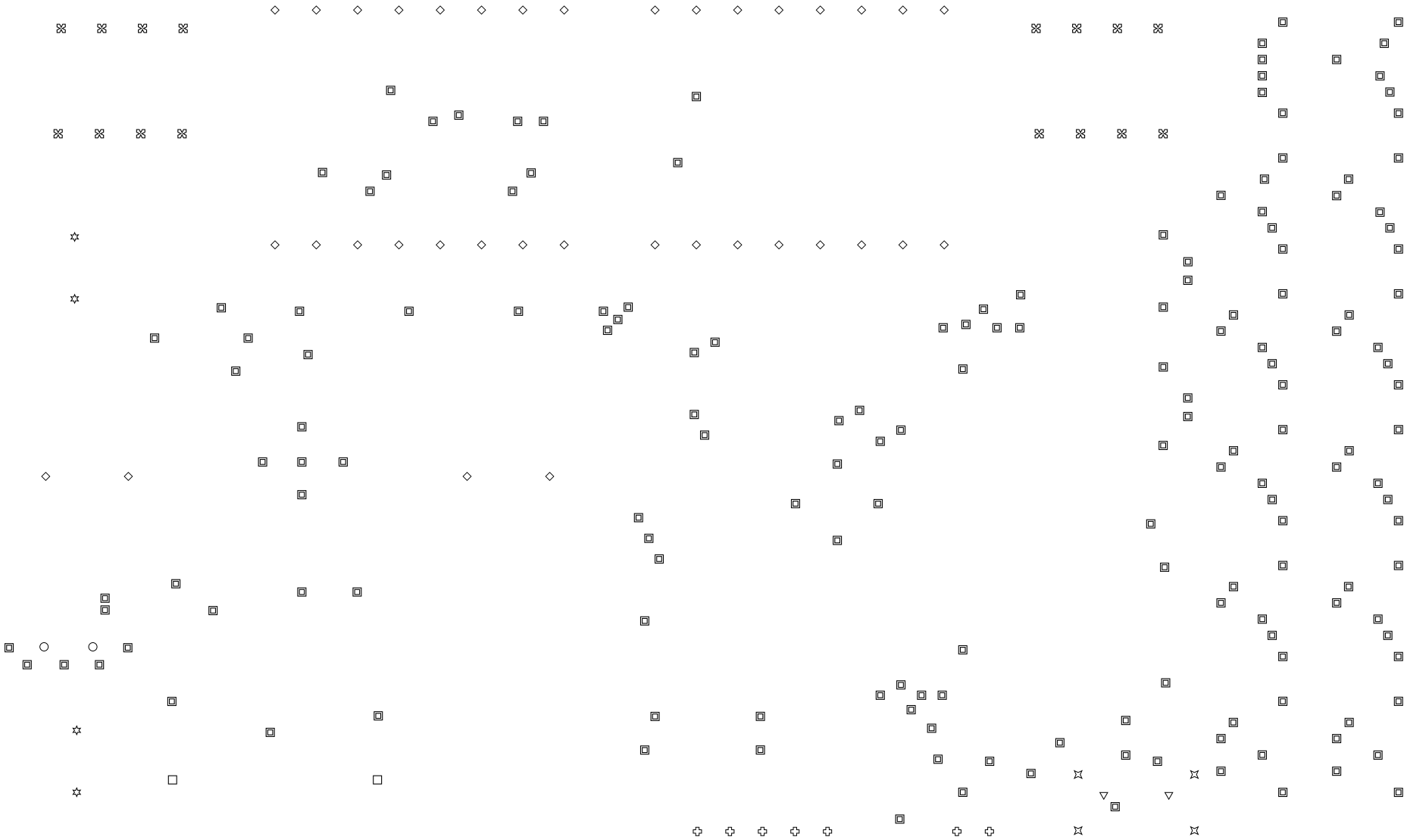

CN3
H2
H5
CN5
CN2
CN6
R11
R12
Q3
R13
Q4
R10
R4
Q1
H3
H6
C5
U2
C7
R6
Q2
CN11
CN7
CN4
D3
H1
H4
R17
R16
SW3
SW2
SW4
C10
R5
C9
C8
CN13
CN10
LED4
R7
C15
C14
C16
R14
R15
C1
U1
C6
U3
CN8
CN9
R3
LED3
CN15
CN14
SW1
C2
C3
L1
D2
LED2
R2
CN16
CN12
R1
LED1
C4
LED5
R8
LED6
R9
C11
C12
U6
CN1
D1
C13
U4
USB1
U5

| No. | Quantity | Comment | Designator | Footprint | Value | Manufacturer Part | Manufacturer | Supplier Part | Supplier |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 470uF | C1 | CAP-TH_BD10.0-P5.00-D1.0-FD | 470uF | GR477M035G15RR0VL4FP0 | CX(承兴) | C45081 | LCSC |
| 2 | 1 | 1uF | C2 | C1206 | 1uF | 1206B105K500CT | Walsin(华新科) | C123655 | LCSC |
| 3 | 7 | 100nF | C3,C4,C9,C13,C14,C15,C16 | C0603 | 100nF | C0603C104J5RAC | KEMET(基美) | C490602 | LCSC |
| 4 | 1 | 1uF | C5 | C0805 | 1uF | CL21B106KAYQNNE | SAMSUNG(三星) |  | LCSC |
| 5 | 1 | 330uF | C6 | CAP-TH_BD10.0-P5.00-D1.0-FD | 330uF | GR337M035G13RR0VL4FP0 | CX(承兴) | C45080 | LCSC |
| 6 | 1 | 4.7uF/50V/X7R | C7 | C0805 | 4.7uF/50V/X7R | CL21B106KAYQNNE | SAMSUNG(三星) |  | LCSC |
| 7 | 1 | 10uF | C8 | C0603 | 10uF | CL10A106MA8NRNC | SAMSUNG(三星) | C96446 | LCSC |
| 8 | 3 | 1uF | C10,C11,C12 | C0603 | 1uF | CGA0603X5R105K500JT | HRE(芯声) | C6119852 | LCSC |
| 9 | 2 | WJ15EDGVC-3.81-2P | CN1,CN4 | CONN-TH_WJ15EDGVC-3.81-2P |  | WJ15EDGVC-3.81-2P | KANGNEX(康奈克斯电气) | C8396 | LCSC |
| 10 | 12 | BM04B-SRSS-TB(LF)(SN) | CN2,CN6,CN7,CN8,CN9,CN10,CN11,CN12,CN13,CN14,CN15,CN16 | CONN-SMD_BM04B-SRSS-TB |  | BM04B-SRSS-TB(LF)(SN) | JST | C160390 | LCSC |
| 11 | 2 | ZX-XH2.54-4PZZ | CN3,CN5 | CONN-TH_4P-P2.50_4PIN |  | ZX-XH2.54-4PZZ | Megastar(兆星) | C7429634 | LCSC |
| 12 | 1 | 1N5822 | D1 | DO-27-BD3.5-L8.6-P12.6-D1.3-L-RD |  | 1N5822 | LGE(鲁光) | C402220 | LCSC |
| 13 | 1 | 1N4148W | D2 | SOD-123_L2.7-W1.6-LS3.7-FD |  | 1N4148W | R+O(宏嘉诚) | C7420318 | LCSC |
| 14 | 1 | SS14 | D3 | SMA_L4.3-W2.6-LS5.2-RD |  | SS14 | onsemi(安森美) | C83852 | LCSC |
| 15 | 4 | B-2100S08P-A110 | H1,H2,H4,H5 | HDR-TH_8P-P2.54-V-M |  | B-2100S08P-A110 | Ckmtw(灿科盟) | C124381 | LCSC |
| 16 | 2 | B-2100S04P-A110 | H3,H6 | HDR-TH_4P-P2.54-V-M |  | B-2100S04P-A110 | Ckmtw(灿科盟) | C124378 | LCSC |
| 17 | 1 | 33uH | L1 | IND-SMD_L12.3-W12.3 | 33uH | SMDRI127-330MT | SXN(顺翔诺) | C9400 | LCSC |
| 18 | 1 | GL0603UR01 | LED1 | LED0603-FD |  | GL0603UR01 | Guiguang light(桂光光电) | C51933292 | LCSC |
| 19 | 3 | XL-1608PGC-06 | LED2,LED3,LED4 | LED0603-RD_GREEN |  | XL-1608PGC-06 | XINGLIGHT(成兴光) | C7371905 | LCSC |
| 20 | 1 | GL0603UB01 | LED5 | LED0603-RD_BLUE |  | GL0603UB01 | Guiguang light(桂光光电) | C51933294 | LCSC |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 21 | 1 | LTST-C191KSKT | LED6 | LED0603-RD-YELLOW | | LTST-C191KSKT | LITEON(光宝) | C125100 | LCSC |
| 22 | 2 | SS8050-G(RANGE:120-200) | Q1,Q2 | SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR | | SS8050-G(RANGE:120-200) | CJ(江苏长电/长晶) | C164886 | LCSC |
| 23 | 1 | S8050 | Q3 | SOT-23-3_L3.0-W1.7-P0.95-LS2.9-BR | | S8050 | MDD(辰达半导体) | C364312 | LCSC |
| 24 | 1 | 2N7002 | Q4 | SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR | | 2N7002 | R+O(宏嘉诚) | C7420321 | LCSC |
| 25 | 5 | 1kΩ | R1,R2,R3,R8,R9 | R0603 | 1kΩ | 0603WAF1001T5E | UNI-ROYAL(厚声) | C21190 | LCSC |
| 26 | 9 | 10kΩ | R4,R5,R6,R10,R12,R14,R15,R16,R17 | R0603 | 10kΩ | 0603WAF1002T5E | UNI-ROYAL(厚声) | C25804 | LCSC |
| 27 | 1 | 1kΩ | R7 | R0603 | 1kΩ | 0603WAF1002T5E | UNI-ROYAL(厚声) | | LCSC |
| 28 | 1 | 510Ω | R11 | R1206 | 510Ω | 1206W4F1000T5E | UNI-ROYAL(厚声) | | LCSC |
| 29 | 1 | 20kΩ | R13 | R0603 | 20kΩ | 0603WAF1002T5E | UNI-ROYAL(厚声) | | LCSC |
| 30 | 1 | MSS12C02LS-BB2.0 | SW1 | SW-SMD_MSS12C02LS-HB2.0-1 | | MSS12C02LS-BB2.0 | SHOU HAN(首韩) | C3008585 | LCSC |
| 31 | 3 | ST-1185S | SW2,SW3,SW4 | SW-SMD_ST-1185S | | ST-1185S | RISHENG(日昇) | C589191 | LCSC |
| 32 | 1 | AP1501S-5.0RG | U1 | TO-263-5_L10.2-W9.9-P1.70-LS14.4-TL | | AP1501S-5.0RG | HTC | C91094 | LCSC |
| 33 | 1 | SPX3819M5-L-3-3/TR | U2 | SOT-23-5_L3.0-W1.7-P0.95-LS2.8-BL | | SPX3819M5-L-3-3/TR | MaxLinear(迈凌) | C9055 | LCSC |
| 34 | 1 | ESP32-S3-WROOM-1-N16R2 | U3 | WIRELM-SMD_ESP32-S3-WROOM-1 | | ESP32-S3-WROOM-1-N16R2 | ESPRESSIF(乐鑫) | C2913205 | LCSC |
| 35 | 1 | ZX-PZ2.0-1-2PZZ | U4 | HDR-TH_2P-P2.00-V-M | | ZX-PZ2.0-1-2PZZ | Megastar(兆星) | C7501287 | LCSC |
| 36 | 1 | ZX-PZ2.0-1-5PZZ | U5 | HDR-TH_5P-P2.00-V-M | | ZX-PZ2.0-1-5PZZ | Megastar(兆星) | C7501286 | LCSC |
| 37 | 1 | CH343P | U6 | TQFN-16_L3.0-W3.0-P0.50-BL-EP1.7 | | CH343P | WCH(南京沁恒) | C2846043 | LCSC |
| 38 | 1 | MICRO 4P DIP | USB1 | MINI-USB-SMD_MICRO-4P-DIP | | MICRO 4P DIP | SHOU HAN(首韩) | C456008 | LCSC |


霍尔阵列
IIC1行（1-6）
IIC2行（7-11）
I019
I020
I03
I046
预留
行列1
行列2
行列3
行列4
行列5
行列6
行列7
行列8
行列9
行列10
行列11
1B 1A 2A 2B
电机1
电机2
驱动1
驱动2
16
1
正
负
电磁铁
定义
关
开
运行
调试
电源
+12V GND
PAS-SET_A0
RXD
TXD
I038 I039 I040 I041 I042
I01 I02